

EXPERIMENT: STUDENT PERFORMANCE ANALYSIS SYSTEM

Experiment No.	02
Date	18/09/2026
Title	Student Performance Analysis System using Java OOP
Real-time Application	College Student Performance Analysis

Objective / Aim

To design and implement a student performance analysis system in Java using Object-Oriented Programming principles to calculate marks, percentage, grade, highest and lowest marks, and overall academic performance.

Problem Statement

Develop a Java application for a college to analyze a student's academic performance. The system accepts student details and marks and calculates total, average, highest, lowest, percentage, grade, pass/fail status, formatted name, and a performance remark.

- Store five subject marks using an array.
- Calculate total and average marks.
- Find highest and lowest marks.
- Calculate percentage and round it to two decimal places.
- Assign grade and determine pass/fail.
- Use trim(), toUpperCase(), and length() on the student's name.
- Display a performance remark.

System Requirements

Hardware: Standard PC/Laptop, 4 GB RAM or more.

Software: Windows/Linux/macOS, JDK 17 or later, Command Prompt/Terminal, and any Java IDE/text editor.

Algorithm / Logic Description

1. Create a class named Student.
2. Declare roll number, student name, and an integer array of marks.
3. Create a constructor to initialize student details.
4. Use a loop to calculate total marks.
5. Use Math.max() and Math.min() to find highest and lowest marks.
6. Calculate average and percentage.
7. Use Math.round(percentage * 100.0) / 100.0 for two decimal places.
8. Process the name using trim(), toUpperCase(), and length().
9. Assign grades: 90–100 A+, 80–89 A, 70–79 B, 60–69 C, 50–59 D, below 50 F.
10. Pass the student when percentage is at least 50%.
11. Generate a performance remark based on the grade.
12. Create a Student object in main() and display the complete report.

Grade Criteria

Percentage	Grade	Result
90–100	A+	PASS
80–89	A	PASS
70–79	B	PASS
60–69	C	PASS

50–59	D	PASS
Below 50	F	FAIL

Java Source Code

```
import java.util.Scanner;

class Student {
    private int rollNumber;
    private String studentName;
    private int[] marks;

    Student(int rollNumber, String studentName, int[] marks) {
        this.rollNumber = rollNumber;
        this.studentName = studentName;
        this.marks = marks;
    }

    int calculateTotal() {
        int total = 0;
        for (int mark : marks) total += mark;
        return total;
    }

    double calculateAverage() {
        return calculateTotal() / (double) marks.length;
    }

    int findHighest() {
        int highest = marks[0];
        for (int mark : marks) highest = Math.max(highest, mark);
        return highest;
    }

    int findLowest() {
        int lowest = marks[0];
        for (int mark : marks) lowest = Math.min(lowest, mark);
        return lowest;
    }

    double calculatePercentage() {
        double p = (calculateTotal() / (marks.length * 100.0)) * 100;
        return Math.round(p * 100.0) / 100.0;
    }

    String determineGrade() {
        double p = calculatePercentage();
        if (p >= 90) return "A+";
        else if (p >= 80) return "A";
        else if (p >= 70) return "B";
        else if (p >= 60) return "C";
        else if (p >= 50) return "D";
        else return "F";
    }

    String getRemark() {
        switch (determineGrade()) {
            case "A+": return "Outstanding Performance";
            case "A": return "Excellent Performance";
            case "B": return "Very Good Performance";
            case "C": return "Good Performance";
            case "D": return "Satisfactory Performance";
            default: return "Needs Improvement";
        }
    }

    void displayDetails() {
        String formattedName = studentName.trim().toUpperCase();

        System.out.println("\n===== STUDENT PERFORMANCE REPORT =====");
        System.out.println("Roll Number      : " + rollNumber);
        System.out.println("Student Name   : " + formattedName);
        System.out.println("Name Length    : " + formattedName.length());

        for (int i = 0; i < marks.length; i++)
            System.out.println("Subject " + (i + 1) + " : " + marks[i]);

        System.out.println("Total Marks      : " + calculateTotal());
        System.out.printf("Average Marks    : %.2f\n", calculateAverage());
        System.out.println("Highest Mark     : " + findHighest());
        System.out.println("Lowest Mark      : " + findLowest());
        System.out.printf("Percentage       : %.2f%%\n", calculatePercentage());
        System.out.println("Grade           : " + determineGrade());
        System.out.println("Result          : " +
            (calculatePercentage() >= 50 ? "PASS" : "FAIL"));
        System.out.println("Performance Remark: " + getRemark());
    }
}

public class StudentPerformance {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
    }
}
```

```
System.out.print("Enter Roll Number: ");
int roll = sc.nextInt();
sc.nextLine();

System.out.print("Enter Student Name: ");
String name = sc.nextLine();

int[] marks = new int[5];

for (int i = 0; i < marks.length; i++) {
    System.out.print("Enter marks for Subject " + (i + 1) + ": ");
    marks[i] = sc.nextInt();
}

Student student = new Student(roll, name, marks);
student.displayDetails();
sc.close();
}
```

Code Explanation

Scanner — Accepts roll number, name and five subject marks from the keyboard.

Student class — Encapsulates student data and all performance calculations.

Array marks — Stores the marks of five subjects.

calculateTotal() — Uses a for-each loop to add all marks.

calculateAverage() — Divides total marks by the number of subjects.

Math.max() / Math.min() — Finds the highest and lowest marks.

calculatePercentage() — Calculates percentage and rounds it to two decimal places using Math.round().

String operations — trim() removes extra spaces, toUpperCase() converts the name to uppercase, and length() finds the number of characters.

determineGrade() — Uses if-else conditions according to the given grade criteria.

getRemark() — Uses switch to display a suitable academic performance remark.

Object creation — new Student(roll, name, marks) creates the Student object.

displayDetails() — Prints the complete student performance report.

Compilation and Execution Commands

Save the program as **StudentPerformance.java**. Open Command Prompt/Terminal in the folder containing the file.

```
javac StudentPerformance.java
java StudentPerformance
```

Command explanation:

- javac StudentPerformance.java — compiles the Java source code and creates the class files.
- java StudentPerformance — executes the compiled program. The class name is used without .java.

Sample Input

```
Enter Roll Number: 101
Enter Student Name:  Arun Kumar
Enter marks for Subject 1: 95
Enter marks for Subject 2: 88
Enter marks for Subject 3: 92
Enter marks for Subject 4: 85
Enter marks for Subject 5: 90
```

Sample Output

```
===== STUDENT PERFORMANCE REPORT =====
Roll Number      : 101
Student Name     : ARUN KUMAR
Name Length      : 10
Subject 1        : 95
Subject 2        : 88
Subject 3        : 92
Subject 4        : 85
Subject 5        : 90
Total Marks      : 450
Average Marks    : 90.00
Highest Mark     : 95
Lowest Mark      : 85
Percentage       : 90.00%
Grade            : A+
Result           : PASS
Performance Remark: Outstanding Performance
```

Result

The Student Performance Analysis System was successfully implemented in Java using a class, object, array, loops, String operations, Math functions, conditional statements, and switch-case. The program successfully calculates and displays the student's complete academic performance report.